Language Models Resist Alignment: Evidence From Data Compression

Jiaming Ji[1*], Kaile Wang[1*], Tianyi Qiu[1*], Boyuan Chen[1*], Jiayi Zhou[1*]
Changye Li[1], Hantao Lou[1], Juntao Dai[12], Yunhuai Liu[3], Yaodong Yang[12†]

[1]Institute for Artificial Intelligence, Peking University
[2]Beijing Academy of Artificial Intelligence
[3]School of Computer Science, Peking University

Abstract

Large language models (LLMs) may exhibit unintended or undesirable behaviors. Recent works have concentrated on aligning LLMs to mitigate harmful outputs. Despite these efforts, some anomalies indicate that even a well-conducted alignment process can be easily circumvented, whether intentionally or accidentally. Does alignment fine-tuning yield have robust effects on models, or are its impacts merely *superficial*? In this work, we make the first exploration of this phenomenon from both theoretical and empirical perspectives. Empirically, we demonstrate the *elasticity* of post-alignment models, *i.e.*, the tendency to revert to the behavior distribution formed during the pre-training phase upon further fine-tuning. Leveraging compression theory, we formally deduce that fine-tuning disproportionately undermines alignment relative to pre-training, potentially by orders of magnitude. We validate the presence of *elasticity* through experiments on models of varying types and scales. Specifically, we find that model performance declines rapidly before reverting to the pre-training distribution, after which the rate of decline drops significantly. Furthermore, we further reveal that *elasticity* positively correlates with the increased model size and the expansion of pre-training data. Our findings underscore the need to address the inherent *elasticity* of LLMs to mitigate their resistance to alignment.[1]

Our Contribution: The *Elasticity* of LLMs

Language models, fine-tuned with perturbations, exhibit an inverse relationship between normalized compression rate changes and dataset volume, akin to that of a series spring system, revealing the *elasticity* of LLMs.

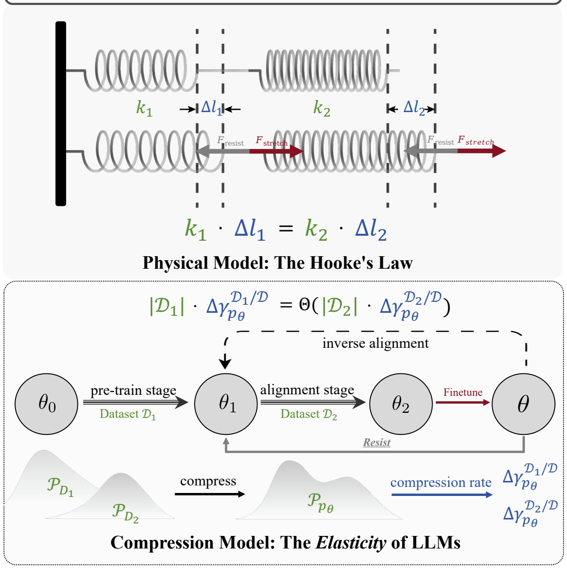

Figure 1: **The *Elasticity* of Language Models.** The change in normalized compression rates ($\Delta\gamma_{p_\theta}^{\mathcal{D}_i/\mathcal{D}}$) and the dataset volume ($|\mathcal{D}_i|$) follows an inverse proportionality law after perturbations, which is akin to the relationship between spring deformation (Δl_i) and stiffness (k_i) in coupled springs. We conjecture that the *elasticity* causes language models to resist alignment, enabling the possibility of *inverse alignment*.

1 Introduction

Large language models (LLMs) have shown remarkable capabilities (Achiam et al., 2023; Zhang et al., 2025). However, due to the inevitable biases and harmful content present in training datasets (Bai et al., 2022a; Ji et al., 2024c; Qian et al., 2024; Lin et al., 2025), LLMs often exhibit behaviors that deviate from human intentions, a phenomenon we refer to as *model misalignment*. Training-based alignment methods, including supervised fine-tuning (SFT), reinforcement learning with human feedback (RLHF) (Ouyang et al., 2022), and other derivatives (Rafailov et al., 2024; Bai et al., 2022b; Lee et al., 2023; Gulcehre et al., 2023; Dong et al., 2023; Xiong et al., 2024; Li et al., 2023; Zhou et al., 2023, 2025; Ji et al., 2024a),

*Equal contribution. †Corresponding author. If you have any questions, feel free to email {jiamg.ji, wkl, tianyi.qiu, cbylll, gaiejj}@stu.pku.edu.cn, yaodong.yang@pku.edu.cn.

[1]The model weight and code are available at `pku-lm-resist-alignment.github.io`.

are the dominant approaches for aligning models. These methods aim to optimize model behavior by rejecting harmful distributions, ensuring LLMs remain consistent with human intentions and values (Ji et al., 2023; Casper et al., 2023).

However, these alignment methods do not truly penetrate the model representations but merely perform *superficial alignment* (Qi et al., 2024a; Cohen et al., 2024; Wen et al., 2024). Recent studies have shown that highly safety-aligned models can become unsafe again with minimal fine-tuning(Yang et al., 2023b; Zhou et al., 2024). Furthermore, fine-tuning aligned LLMs on non-malicious datasets may also weaken models' safety mechanisms (Qi et al., 2024a; Jain et al., 2023).

Why is alignment so fragile?

In this work, we make the first exploration of the possible mechanism behind the counterintuitive phenomenon: the existence of an alignment resistance mechanism in language models. This mechanism may limit the alignment process of LLMs to superficial adjustments. It could allow the reversal or revocation of alignment through a series of technical measures, a concept we refer to as *inverse alignment*. What drives language models to resist alignment? How does this mechanism lead to *inverse alignment*? Our key contributions are summarized as follows:

- **(Phenomenon)** We uncover that language models exhibit *elasticity*, as illustrated in Figure 1 and Theorem 4.2. It encompasses *resistance*: pre-trained models tend to retain their original distribution; and *rebound*: the deeper alignment of models, the faster they return to the pre-trained distribution under reverse finetuning. Moreover, The model's change in compression rates $\Delta\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_i/\mathcal{D}}$ across different datasets is inversely proportional to their sizes $|\mathcal{D}_i|$, which is analogous to the deformation behavior of a series of springs, as illustrated in Section 4.3.

- **(Mechanism)** We systematically model the training and alignment process of language models through compression theorem, as detailed in Section 3.2. We elaborate on the compression protocol of language models to explore their training and alignment processes, laying a foundation for subsequent research on *elasticity*.

- **(Validation)** We experimentally observe consistent *resistance* and *rebound* phenomena across various LLMs, as detailed in Section 5. This highlights the universality of *elasticity* and the need for systematic approaches to achieve robust and deep alignment.

2 Related Work

The Fragility of LLMs Alignment Pre-trained LLMs often generate offensive content. Recent initiatives (Ouyang et al., 2022; Bai et al., 2022a; Yang et al., 2024) have aimed to align these models to minimize harmful outputs (Yang et al., 2023a; Cui et al., 2024). However, research indicates that even well-aligned models can be compromised easily, and fine-tuning them on non-malicious datasets might unintentionally impair their safety mechanisms (Yang et al., 2023b; Qi et al., 2024b; Hubinger et al., 2024; Dong et al., 2024). Moreover, recent study reveals that models selectively adhere to training objectives during the training phase to preserve their inherent preferences (Greenblatt et al., 2024; Lang et al., 2024; Park et al., 2024). Why is alignment so fragile? Wei et al. (2024) use weight attribution to separate safety-critical and utility-related regions at both neuron and rank levels. Qi et al. (2024a) propose the concept of shallow safety alignment, arguing that safety alignment should reach beyond surface-level tokens to shape the model's internal mechanisms.

3 What is *Elasticity*?

We discover that language models exhibit elasticity, which leads to resistance to alignment. In this section, we formally introduce the definition of *elasticity*, along with the compression theory tools used in the analysis. Firstly, we review the training alignment objective and the compression theorem.

3.1 Preliminaries

Pre-training. During pre-training, an LLM acquires foundational language comprehension and reasoning abilities by processing vast quantities of unstructured text. The pre-train loss $\mathcal{L}_{\text{PT}}(\boldsymbol{\theta};\mathcal{D}_{\text{PT}})$ is defined as follows:

$$\mathcal{L}_{\text{PT}}(\boldsymbol{\theta};\mathcal{D}_{\text{PT}}) = -\mathbb{E}_{(\boldsymbol{x},x_N)\sim\mathcal{D}_{\text{PT}}}\left[\log p_{\boldsymbol{\theta}}\left(x_N|\boldsymbol{x}\right)\right],$$

where $\boldsymbol{x} = (x_0, \cdots, x_{N-1})$ and $N \in \mathbb{N}$, such that $(x_0, \cdots, x_N)$ forms a prefix in some piece of pre-training text. $\mathcal{D}_{\text{PT}}$ stands for pre-training dataset.

Supervised Fine-tuning (SFT). SFT adjusts the pre-trained models to follow specific instructions, utilizing a smaller dataset compared to the pre-training corpus to ensure model alignment with target tasks. For $\mathcal{D}_{\text{SFT}} = \left\{\left(\boldsymbol{x}^i, \boldsymbol{y}^i\right)\right\}_{i=1}^{N}$ sampled from a high-quality distribution, SFT aims to minimize the negative log-likelihood loss:

$$\mathcal{L}_{\text{SFT}}(\boldsymbol{\theta}; \mathcal{D}_{\text{SFT}}) = -\mathbb{E}_{(\boldsymbol{x},\mathbf{y})\sim\mathcal{D}_{\text{SFT}}} \left[\log p_{\boldsymbol{\theta}} \left(\boldsymbol{y}|\boldsymbol{x}\right)\right].$$

Given that $\mathbb{E}_{(\boldsymbol{x},\mathbf{y})\sim\mathcal{D}_{\text{SFT}}} \left[\log p_{\mathcal{D}} \left(\boldsymbol{y}|\boldsymbol{x}\right)\right]$ is fixed when specifying $\mathcal{D}_{\text{SFT}}$, the optimization objective $\mathcal{L}_{\text{SFT}}$ becomes the Kullback-Leibler (KL) divergence between $p_{\boldsymbol{\theta}}$ and the SFT distribution.

Lossless Compression. The goal of lossless compression is to find a compression protocol that encodes a given dataset $\mathcal{D}$ and its distribution $\mathcal{P}_{\mathcal{D}}$ with the smallest possible expected length and allows for a decoding scheme that can perfectly reconstruct the original dataset. According to Shannon's source coding theorem (Shannon, 1948), for a random variable follows $\mathcal{P}_{\mathcal{D}}$, the expected code length $\mathcal{L}$ of any lossless compression protocol satisfies:

$$\mathcal{L} \geq H\left(\mathcal{P}_{\mathcal{D}}\right),$$

where $H\left(\mathcal{P}_{\mathcal{D}}\right)$ is the Shannon entropy of $\mathcal{P}_{\mathcal{D}}$.

Compression and Prediction. Compression and prediction are tightly interconnected. Consider a model $p_{\boldsymbol{\theta}}$ and $\boldsymbol{x} = (x_0, \cdots, x_{m-1})$ derived from a dataset $\mathcal{D}$, the expected code length $\mathcal{L}$ under arithmetic coding (Witten et al., 1987) satisfies:

$$\mathcal{L} = \mathbb{E}_{\boldsymbol{x}\sim\mathcal{D}} \left[\sum_{0\leq k\leq m} -\log_2 p_{\boldsymbol{\theta}} \left(x_k | x_{0,\cdots,k-1}\right)\right],$$

which is the current training objective of language models. Minimizing log-likelihood loss is equivalent to minimizing the compression rate when models act as a lossless compressor. Thus, optimal compression and prediction are equivalent (Delétang et al., 2023; Hutter, 2005). Experiments show the equivalence between large language model prediction and compression (Delétang et al., 2023), and that compression performance correlates linearly with intelligence (Huang et al., 2024).

3.2 The Compression Protocol of LLMs

We aim to study the dynamic changes in language models during training and alignment. Given the equivalence between language model training and data compression (Delétang et al., 2023), a feasible modeling approach is to treat the language model as a lossless compression protocol. The process of training and aligning the model on different datasets can be equivalently viewed as the joint compression of these datasets by the protocol. The model's compression rate on various datasets serves as a surrogate metric for the loss during training and alignment. In this section, we detail the modeling specifics of this compression protocol.

Considering the impact of tokenization on compression rate, we use tokenized sequences as input and output modalities. For simplicity, we assume the tokenized vocabulary consists of only binary tokens (specifically `0/1`).

Definition 3.1 (Token Tree $\mathcal{T}$)**.** For a dataset $\mathcal{D} = \{\boldsymbol{z}_i \in \{0|1\}^\infty \mid i = 1, 2, \cdots\}$, the token tree of $\mathcal{D}$, denoted as $\mathcal{T}_{\mathcal{D}}$, is defined as follows: each node has child nodes labeled `0` or `1`, along with an end-of-sequence (EOS) leaf node. The path from the root to a leaf node defines each response $\boldsymbol{z}_i$, with the corresponding EOS node weight representing the response's probability while the weight of non-leaf nodes is the sum of the weights of their child nodes.

In token tree modeling, the model's training process is conceptualized as learning the node weights of the token tree. However, due to the finite size of the model's parameters in practical scenarios, the model cannot accurately capture node weights at arbitrary depths of the token tree. Therefore, we propose the following assumption.

Assumption 3.2 (Scale of $\mathcal{T}$ is Monotone with Model Size)**.** Consider a parameterized model $p_{\boldsymbol{\theta}}(\cdot)$ and a dataset $\mathcal{D}$, We assume that the depth of the portion of $\mathcal{T}_{\mathcal{D}}$ that can be perfectly modelled by $p_{\boldsymbol{\theta}}$ is monotonically increasing with the size of $\boldsymbol{\theta}$.

Based on the above assumption, we can define the compression protocol during the model training and alignment process.

Definition 3.3 (The Compression Protocal)**.** Consider using the model $p_{\boldsymbol{\theta}}(\cdot)$ to compress the dataset $\mathcal{D}$. The compression protocol is defined in two steps: a) Prune the token tree of $\mathcal{D}$, retaining only the top d layers. b) Apply Huffman coding (Huffman, 1952) to compress the pruned token tree. Specifically, each response from the root node to a leaf node is treated as a symbol in the Huffman coding alphabet, and the weight of the leaf node is the probability of the symbol.